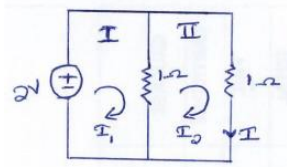
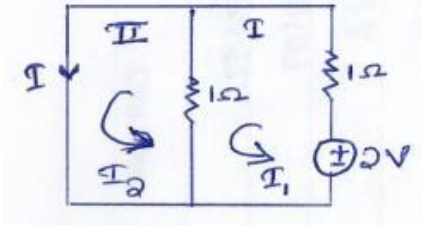
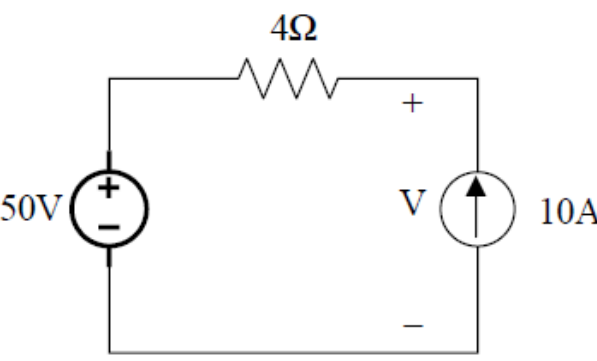


Sahrdaya College of Engineering and Technology, Kodakara		
Answer Key		
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY		
THIRD SEMESTER B. TECH REGULAR AND SUPPLEMENTARY EXAMINATION, DECEMBER 2022 (2019 Scheme)		
Course Code: EET201		
Course Name: CIRCUITS AND NETWORKS		
PART A		
		<i>Answer all questions. Each question carries 3 marks</i>
	Q1	State and explain reciprocity theorem using an example.
	Ans	In a linear, bilateral, active, single source network, the ratio of excitation to response is a constant when their positions are interchanged.  Applying mesh in I $2 - 1(I_1 - I_2) = 0$ $-I_1 + I_2 = -2$ $I_1 - I_2 = 2 \text{ ----- (1)}$ Applying mesh in II $-1(I_2 - I_1) - I_2 = 0 \text{ ----- (2)}$ $-I_2 + I_1 - I_2 = 0$ $I_1 - 2I_2 = 0 \text{ ----- (3)}$ Solving, $I_1 - I_2 = 2$ $I_1 - 2I_2 = 0$ $I_1 = 4 \quad I_2 = 2$ $\therefore I = I_2 = 2$ $\therefore E/I = 2/2 = 1$

		 Mesh I, $2 - 1(I_1) - 1(I_1 - I_2) = 0$ $2 - I_1 - I_1 + I_2 = 0$ $2 - 2I_1 + I_2 = 0$ $2I_1 - I_2 = 2 \text{ ----- (1)}$ Mesh II $-1(I_2 - I_1)$ $-I_2 + I_1 = 0$ $I_1 - I_2 = 0 \text{ ----- (2)}$ $I_1 = 2$ $I_2 = 2$ $I_2 = I = 2$ $\therefore E/I = 2/2 = 1$ Hence reciprocity theorem is verified.
Q2		For the circuit given below, find the voltage 'V' across the 10 A source, using superposition principle. 
Ans		50V source acting alone, $V' = 50V$ 10A source acting alone, $V'' = 40V$ Voltage across the 10A source, $V = V' + V'' = 90V$
Q3		Derive an expression for the circuit in a series RL circuit connected to a DC source.

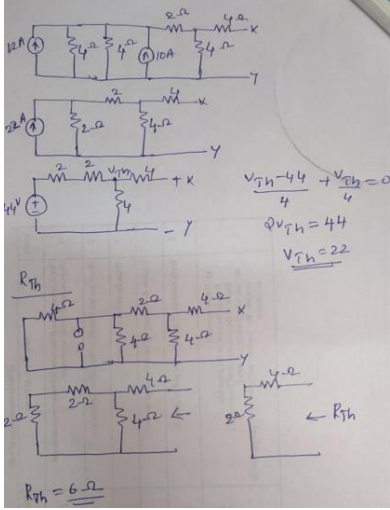
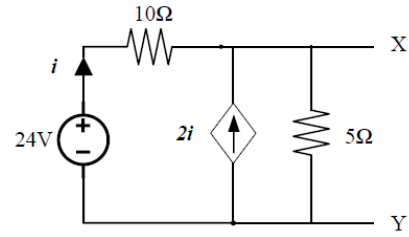
Ans	In a series RL (Resistor-Inductor) circuit connected to a DC source, the circuit consists of a resistor (R), an inductor (L), and a DC voltage source (V). The circuit diagram is shown below: Fig. Using Kirchhoff's voltage law (KVL), we can write: $V - IR - L(di/dt) = 0$ where, • V is the DC voltage source • I is the current through the circuit • R is the resistance of the resistor • L is the inductance of the inductor • di/dt is the rate of change of current through the inductor with respect to time. We can rearrange this equation to obtain an expression for the current: $di/dt = (V - IR)/L$ This is a first-order differential equation, which can be solved using the method of integrating factors. We multiply both sides of the equation by the integrating factor $e^{(Rt/L)}$, where R = resistance of the resistor and L = inductance of the inductor. This gives: $e^{(Rt/L)} di/dt + (R/L)e^{(Rt/L)} i = (V/L) e^{(Rt/L)}$ The left-hand side of the equation can be simplified using the product rule of differentiation and the fact that $d/dt(e^{(Rt/L)} i) = e^{(Rt/L)} di/dt + (R/L) e^{(Rt/L)} i$: $d/dt(e^{(Rt/L)} i) = (V/L) e^{(Rt/L)}$ Integrating both sides with respect to time gives: $e^{(Rt/L)} i = (V/R) e^{(Rt/L)} + C$ where C is the constant of integration. We can solve for the constant C using the initial condition that the current at time $t = 0$ is zero: $e^{(0/L)} i(0) = (V/R) e^{(0/L)} + C$ $C = - (V/R)$ Substituting the value of C into the previous equation gives:
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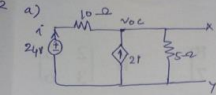
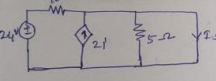
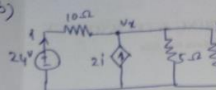
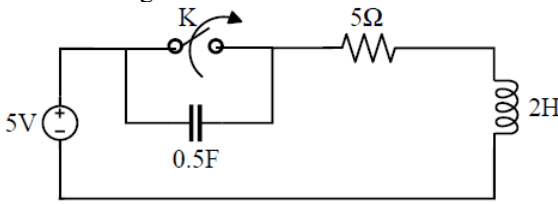
	$i(t) = (V/R) (1 - e^{(-Rt/L)})$ This is the expression for the current in a series RL circuit connected to a DC source. The voltage across the resistor is $V_r = IR$, and the voltage across the inductor is $V_l = L(di/dt)$, which can also be expressed as $V_l = V - V_r$. Therefore, we can write: $V_r = R i = R (V/R) (1 - e^{(-Rt/L)}) = V (1 - e^{(-Rt/L)})$ $V_l = V - V_r = V e^{(-Rt/L)}$ These are the expressions for the voltage across the resistor and the voltage across the inductor in a series RL circuit connected to a DC source.
Q4	Define time constant of a circuit. A series RC circuit is connected to a DC source of 10V at time $t=0$. Find the voltage across the capacitor at $t=1s$, if the time constant of the circuit is 2 seconds.
Ans	Time Constant is the time taken to reach the final steady state value. $V = 10V \Rightarrow t=0$ V at $t=1s$ time constant = 2 seconds $\Rightarrow RC$ $V_c(t) = V (1 - e^{-t/RC})$ $V_c(t) = 10 (1 - e^{-1/2}) = \underline{\underline{3.93V}}$
Q5	Define transfer function of a network. Derive the transfer function of a series RLC circuit by taking the current through the resistor as the output.
Ans	The transfer function of a network is a mathematical function that describes the relationship between the input and output signals of the network. It represents the frequency response of the network, i.e., how the network responds to different frequencies of input signals. A series RLC circuit consists of a resistor, an inductor, and a capacitor connected in series with each other. The current through the resistor can be taken as the output signal of the circuit. To derive the transfer function of this circuit, we first need to write the differential equation governing the behavior of the circuit. The voltage across the resistor is given by Ohm's law as $V_R = RI$, where I is the current through the resistor. The voltage across the inductor and capacitor are given by the equations $V_L = L(di/dt)$ and $V_C = 1/C \int I dt$, respectively, where L is the inductance and C is the capacitance of the circuit. Applying Kirchhoff's voltage law, we get the differential equation: $V_R + V_L + V_C = 0$

		Substituting the above expressions for V_R, V_L, and V_C, we get: $RI + L(dI/dt) + 1/C \int I dt = 0$ Taking the Laplace transform of both sides, we get: $RI(s) + LsI(s) + 1/C(1/s)I(s) = 0$ Simplifying the above equation, we get: $I(s) = (s/(RCs^2 + L*s + 1))V_{in}(s)$ where $V_{in}(s)$ is the Laplace transform of the input voltage. Therefore, the transfer function $H(s)$ of the series RLC circuit with the current through the resistor as the output is: $H(s) = I(s)/V_{in}(s) = s/(RCs^2 + L*s + 1)$ This transfer function describes how the circuit responds to different input frequencies, and can be used to analyze and design the circuit for specific performance criteria.
	Q6	Explain the use of dot convention in the analysis of coupled circuits.
	Ans	The dot convention is a method used to represent the direction of current flow and polarity of voltage across a transformer or a coupled circuit. This convention is important because it helps to avoid errors when analyzing the circuit. In the dot convention, a dot is placed at one end of each winding of a transformer or a coupled circuit. The dot indicates the direction of the current that enters the winding. The other end of the winding is assumed to be connected to a common node or a reference point. When two coupled circuits are connected, the dots are aligned in a specific manner. If the dots of the two coupled circuits are on the same side, then the coupling is said to be in-phase, and if the dots are on opposite sides, the coupling is said to be out-of-phase. Using the dot convention, we can determine the polarity of the voltage induced in the secondary winding of a transformer or a coupled circuit. If the current flows from the dot end to the other end of the primary winding, then the induced voltage in the secondary winding will have the same polarity as the primary voltage. If the current flows from the other end to the dot end of the primary winding, then the induced voltage in the secondary winding will have the opposite polarity as the primary voltage. The dot convention is also useful in determining the direction of energy flow in a coupled circuit. If the dots of the two coupled circuits are in-phase, then energy flows from the primary circuit to the secondary circuit. If the dots are out-of-phase, then energy flows from the secondary circuit to the primary circuit.

		Overall, the dot convention simplifies the analysis of coupled circuits by providing a standard way to represent the direction of current flow and voltage polarity. This reduces confusion and helps to avoid errors when analyzing the circuit.
	Q7	Determine the current through the neutral wire in a three phase 4 wire system, if the phase currents of the load are $I_R = 10\angle -30^\circ A$, $I_Y = 5\angle -50^\circ A$ and $I_B = 5\angle 50^\circ A$.
	Ans	Neutral current, $I_N = I_R + I_Y + I_B$ $I_N = 15.9\angle -18.3^\circ A$
	Q8	A series RLC circuit with $R = 10\Omega$, $L = 2H$ and $C = 40\mu F$ is connected to a variable frequency AC supply. Determine the frequency of the supply, for which the phase angle between the circuit current and supply voltage is zero.
	Ans	Resonant frequency, $f_r = \frac{1}{2\pi\sqrt{LC}}$ $f_r = 17.8 \text{ Hz}$ or $\omega_r = 111.8 \text{ rad/s}$
	Q9	Define h parameters of a two-port network. Why are they called hybrid parameters?
	Ans	The h-parameters (also known as hybrid parameters or ABCD parameters) are a set of four parameters that describe the behavior of a two-port network. The h-parameters are defined as follows: h_{11} = input impedance looking into port 1 with port 2 open-circuited h_{12} = reverse voltage gain (the change in voltage at port 1 for a given change in current at port 2), with port 1 open-circuited h_{21} = forward current gain (the change in current at port 2 for a given change in voltage at port 1), with port 2 open-circuited h_{22} = output impedance looking into port 2 with port 1 short-circuited These parameters are typically represented in matrix form as follows: $\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$ The h-parameters are called "hybrid" because they allow us to model the behavior of a two-port network as a combination of voltage and current sources. Specifically, h_{11} and h_{22} represent impedances, which can be modeled as voltage sources in series with a resistor, while h_{12} and h_{21} represent gains, which can be modeled as current sources in parallel with a resistor. This hybrid model makes it easy to analyze and design two-port networks for a wide range of applications.
	Q10	Obtain Y parameters of a two-port network whose Z parameters are given by $z_{11} = 4\Omega$, $z_{12} = 2\Omega$, $z_{21} = 3\Omega$ and $z_{22} = 4\Omega$.

Ans	$[Y] = [Z]^{-1}$ $\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 3 & 4 \end{bmatrix}$ $\Delta Z = Z_{11} Z_{22} - Z_{12} Z_{21}$ $= 4 \times 4 - 0 \times 3 = 16 - 0 = \underline{16}$ $Y_{11} = \frac{Z_{22}}{\Delta Z} = \frac{4}{16} = 0.25$ $Y_{12} = \frac{-Z_{12}}{\Delta Z} = \frac{-0}{16} = -0.0$ $Y_{21} = \frac{-Z_{21}}{\Delta Z} = \frac{-3}{16} = -0.1875$ $Y_{22} = \frac{Z_{11}}{\Delta Z} = \frac{4}{16} = 0.25$ $[Y] = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} 0.25 & -0.0 \\ -0.1875 & 0.25 \end{bmatrix}$
PART B	
<i>Answer any one full question from each module. Each question carries 14 marks</i>	
Module 1	
Q11	For the circuit given below, - Determine the Thevenin's equivalent circuit across the terminal X and Y. - Determine the value of resistance to be connected across X and Y so that maximum power is transferred to it. Also, calculate the maximum power transferred.
Ans	a)

		 $V_{Th} = 22$ $R_{Th} = 6\Omega$ b) Max power transfer $R_L = R_{Th} = 6\Omega$ $P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{22^2}{4 \times 6} = 20.16W$
Q12	For the circuit given below, a) Find the Norton's equivalent circuit across the terminals X and Y. b) If a 10Ω resistor is connected across the terminals X and Y, find the power dissipated in it.	
Ans	a)	

		2 a)  $\frac{V_{OC} - 24}{10} - 2i + \frac{V_{OC}}{5} = 0$ $\frac{V_{OC} - 24}{10} - 2\left(\frac{V_{OC} - 24}{10}\right) + \frac{V_{OC}}{5} = 0$ $\frac{3(V_{OC} - 24)}{10} + \frac{V_{OC}}{5} = 0$ $3(V_{OC} - 24) + 2V_{OC} = 0$ $5V_{OC} = 24 \times 3 \Rightarrow V_{OC} = 14.4V$  $3i = I_{SC}$ $i = \frac{24}{10} = 2.4 \quad I_{SC} = 3 \times 2.4 = 7.2$ $R_N = \frac{V_{OC}}{I_{SC}} = \frac{14.4}{7.2} = 2\Omega$ b) 3)  $\frac{V_X - 24}{10} - 2i + \frac{V_X}{5} + \frac{V_X}{10} = 0$ $\frac{V_X - 24}{10} + 2\left(\frac{V_X - 24}{10}\right) + \frac{V_X}{5} + \frac{V_X}{10} = 0$ $\frac{3(V_X - 24) + V_X + 2V_X}{10} = 0 \Rightarrow 3V_X - 72 + 3V_X = 0$ $6V_X = 72$ $V_X = 12$ $\frac{V_X}{10} = \frac{12}{10} = 1.2V$
Module 2		
Q13	The switch K in the circuit given below has been in the closed position for a long time and steady state condition is reached. At $t = 0$, the switch is opened. Find the expression for the current through the inductor for $t > 0$. 	

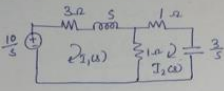
Ans

3)

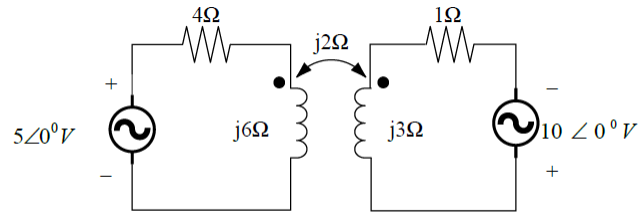
$$\begin{aligned}
 & -5I(t) - 2.5I(t) + 2 + \frac{5}{s} - \frac{1}{0.5s} I(t) \\
 & -I(t) \left(5 + 2.5 + \frac{1}{0.5s} \right) = \frac{-2.5 - 5}{s} \\
 & I(t) \left(\frac{s^2 + 2.5s + 1}{0.5s} \right) = \frac{2.5 + 5}{s} \\
 & I(t) = \frac{2.5 + 5}{s} \times \frac{0.5s}{s^2 + 2.5s + 1} \\
 & I(t) = \frac{5 + 2.5}{(s + \frac{1}{2})(s + 2)} \\
 & I(t) = \frac{A}{(s + \frac{1}{2})} + \frac{B}{(s + 2)} \\
 & \frac{5 + 2.5}{(s + \frac{1}{2})(s + 2)} = \frac{A(s + 2) + B(s + \frac{1}{2})}{(s + \frac{1}{2})(s + 2)} \\
 & 5 + 2.5 = As + 2A + Bs + \frac{B}{2} \\
 & 2A + \frac{B}{2} = 2.5 \\
 & A + B = 1 \quad A = \frac{4}{3}, B = -\frac{1}{3} \\
 & I(t) = \frac{4/3}{(s + 1/2)} + \frac{-1/3}{(s + 2)} \\
 & i(t) = \left[\frac{4}{3} e^{-1/2t} - \frac{1}{3} e^{-2t} \right] A
 \end{aligned}$$

Q14 The switch K in the circuit given below was initially at position 1 and the circuit has been at steady state condition. At time $t = 0$, the switch is moved to position 2. Find the expression for the current through the inductor for $t > 0$.

	Ans	KVL equations! $3i_1 - 2i_2 = 5$ $-2i_1 + 4i_2 + 2 \frac{di_2}{dt} = 0$ Differential equation for $t > 0$ $\frac{di_2}{dt} + \frac{4}{3}i_2 = \frac{5}{3}$ Circuit at $t = 0^-$ $i_2(0^-) = 2.5A$ $i_2(t) = \frac{5}{3} + 2.5e^{-\frac{4}{3}t}$
Module 3		
	Q15	In the network given below, the inductor and capacitor are initially relaxed. The switch K is closed at time $t = 0$. - Model the circuit in s-domain for $t > 0$ - Using mesh analysis, determine the expression for the current through the inductor for $t > 0$.
	Ans	a)

		a) Transformed circuit  b) Applying KVL in mesh 1 $\frac{10}{s} - 3I_1(s) - sI_1(s) - (I_1(s) + I_2(s)) = 0$ $\frac{10}{s} - 3I_1(s) - sI_1(s) - I_1(s) + I_2(s) = 0$ $\frac{10}{s} - (4+s)I_1(s) + I_2(s) = 0$ $(s+4)I_1(s) - I_2(s) = \frac{10}{s} \quad (1)$ mesh 2 $-I_2(s) - \frac{3}{s}I_2(s) - [I_2(s) - I_1(s)] = 0$ $-I_2(s) - \frac{3}{s}I_2(s) - I_2(s) + I_1(s) = 0$ $I_1(s) - [2 + \frac{3}{s}]I_2(s) = 0$ $-I_1(s) + (2 + \frac{3}{s})I_2(s) = 0 \quad (2)$ $\begin{bmatrix} s+4 & -1 \\ -1 & 2+\frac{3}{s} \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} \frac{10}{s} \\ 0 \end{bmatrix}$ $I_1(s) = \frac{\begin{vmatrix} \frac{10}{s} & -1 \\ 0 & 2+\frac{3}{s} \end{vmatrix}}{\begin{vmatrix} s+4 & -1 \\ -1 & 2+\frac{3}{s} \end{vmatrix}} = \frac{\frac{10}{s} [2 + \frac{3}{s}]}{(s+4)(2 + \frac{3}{s}) - 1}$ $= \frac{\frac{10}{s} [2 + \frac{3}{s}]}{2s + 3 + 8 + \frac{12}{s} - 1} = \frac{\frac{10}{s} [2 + \frac{3}{s}]}{2s + \frac{12}{s} + 10}$ $= \frac{\frac{10}{s} [2s + 3]}{\frac{2s^2 + 12 + 10s}{s}} = \frac{10 [2s + 3]}{s(2s + 10s + 12)}$ $I_1(s) = \frac{5(2s+3)}{s(s+3)(s+2)}$ Partial fraction method $I_1(s) = \frac{A}{s} + \frac{B}{s+3} + \frac{C}{s+2}$ $A = I_1(s) \times s \Big _{s=0} = 2.5$ $B = I_1(s) \times (s+3) \Big _{s=-3} = -5$ $C = I_1(s) \times (s+2) \Big _{s=-2} = 2.5$ $I_1(s) = \frac{2.5}{s} - \frac{5}{s+3} + \frac{2.5}{s+2}$ $i_1(t) = \underline{2.5 - 5e^{-3t} + 2.5e^{-2t}}$ b) Mesh equations: $\begin{bmatrix} s+4 & -1 \\ -1 & 2+\frac{3}{s} \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} \frac{10}{s} \\ 0 \end{bmatrix}$ $I_1(s) = \frac{10(s+1.5)}{s(s+2)(s+3)}$ $i_1(t) = 2.5 + 2.5e^{-2t} - 5e^{-3t}$	
Q16	For the circuit given below, a) Find the steady state current through the 1Ω and 4Ω resistors.		

b) Obtain the conductively coupled equivalent circuit.



Ans

a)

Here M is $j2\Omega$ $\therefore I_1$ is entering the dot up I_2 is leaving the dot $M = -j2\Omega$

Applying KVL in loop ①

$$4I_1 + j6I_1 - j2I_2 = 5\angle 0^\circ V$$

$$(4+j6)I_1 - j2I_2 = 5\angle 0^\circ V \quad \text{--- ①}$$

Applying KVL in loop ②

$$j3I_2 + I_2 - j2I_1 = 10\angle 0^\circ V$$

$$-j2I_1 + (3+j3)I_2 = 10\angle 0^\circ V \quad \text{--- ②}$$

$$\begin{bmatrix} 5\angle 0 \\ 10\angle 0 \end{bmatrix} = \begin{bmatrix} 4+j6 & -j2 \\ -j2 & 1+j3 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$$I_1 = \frac{\begin{vmatrix} 5\angle 0 & -j2 \\ 10\angle 0 & 1+j3 \end{vmatrix}}{\Delta} \quad \Delta = \begin{vmatrix} 4+j6 & -j2 \\ -j2 & 1+j3 \end{vmatrix}$$

$$= -10 + 18j$$

$$= \frac{(5\angle 0)(1+j3) - (j2 \times 10\angle 0)}{-10 + 18j}$$

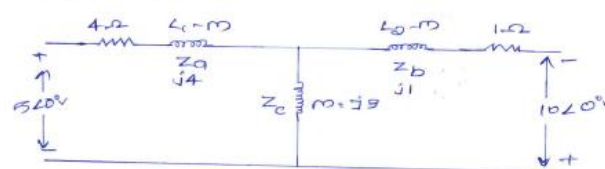
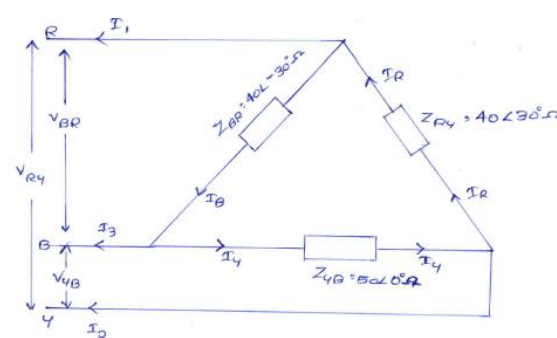
$$= 1.367 + 1.037j \approx 1.37 - 1.04j A$$

$$= \underline{\underline{1.37 - j1.04 A}}$$

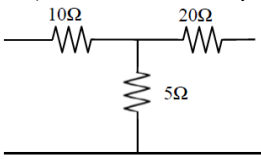
$$I_2 = \frac{\begin{vmatrix} 4+j6 & 5\angle 0 \\ -j2 & 10\angle 0 \end{vmatrix}}{-10 + 18j} = 2.028 - 3.349j$$

$$\approx \underline{\underline{2.03 - j3.35 A}}$$

b)

		Here m is -ve if I_1 & I_2 are opposite in direction. Hence case 2:  $Z_a = L_1 - m = j6 - j3$ $Z_b = L_2 - m = j3 - j3$ $Z_c = m = j3$ $Z_a = j4 \quad Z_b = j1 \quad Z_c = j3$ $V_1 = (L_1 - m) \frac{di_1}{dt} + m \left(\frac{d}{dt} (i_1 - i_2) \right)$ $V_2 = (L_2 - m) \frac{di_2}{dt} + m \left(\frac{d}{dt} (i_2 - i_1) \right)$
Module 4		
Q17		A 400V, three-phase balanced supply feeds a delta-connected load having phase impedances $Z_{RY} = 40 \angle 30^\circ \Omega$, $Z_{YB} = 50 \angle 0^\circ \Omega$ and $Z_{BR} = 40 \angle -30^\circ \Omega$. Determine the following. - Phase currents - Line currents - Active and reactive power delivered to the load
Ans		 Here $V = 400V$ $Z_{RY} = 40 \angle 30^\circ \Omega = 34.64 + j20 \Omega$ $Z_{YB} = 50 \angle 0^\circ \Omega = 50 + j0 \Omega$ $Z_{BR} = 40 \angle -30^\circ \Omega = 34.64 - j20 \Omega$ $V_{RY} = V \angle 0^\circ = 400 \angle 0^\circ V$ $V_{YB} = V \angle -120^\circ = 400 \angle -120^\circ V$ $V_{BR} = V \angle -240^\circ = 400 \angle -240^\circ V$

	<u>Phase Currents</u> $I_R = \frac{V_{R4}}{Z_{R4} \angle \phi_1} = \frac{400 \angle 0}{40 \angle 30} = 10 \angle -30^\circ \text{ A}$ $I_Y = \frac{V_{YB}}{Z_{YB} \angle \phi_1} = \frac{400 \angle -120}{50 \angle 40} = 8 \angle -100^\circ \text{ A}$ $I_B = \frac{V_{BR}}{Z_{BR} \angle \phi_2} = \frac{400 \angle -240}{40 \angle -30} = 10 \angle 150^\circ \text{ A}$ <u>Line Currents</u> $I_1 = I_R - I_B = (10 \angle -30) - (10 \angle 150) = 20 \angle -30^\circ \text{ A}$ $I_2 = I_Y - I_R = (8 \angle -100) - (10 \angle -30) = 12.80 \angle -171.94^\circ \text{ A}$ $I_3 = I_B - I_Y = (10 \angle 150) - (8 \angle -100) = 10.80 \angle 111.34^\circ \text{ A}$ Power Delivered to R-Phase: $(I_R)^2 \times R_R$ $= 10^2 \times 34.64 = 3464 \text{ W}$ Power Delivered to Y Phase: $(I_Y)^2 \times R_Y$ $= 8^2 \times 50 = 3200 \text{ W}$ Power Delivered to B Phase: $(I_B)^2 \times R_B$ $= 10^2 \times 34.64 = 3464 \text{ W}$ Power delivered to the load = $3464 + 3200 + 3464$ $= 10128 \text{ W}$
Q18	A resistor, a capacitor and an inductor are connected in series with a 230 V, variable frequency AC source. When the supply frequency is varied to 25Hz, the circuit offered a minimum impedance of 50Ω. If the inductance of the circuit is 1H, determine the following. - Resistance and capacitance of the circuit. - The voltage across the capacitor. - The supply frequencies at which the power dissipated in the resistor is half that of at 25Hz. - Q factor at resonance and bandwidth of the circuit.
Ans	① At Resonance, $Z = R = 50\Omega$ Resistance, $R = 50\Omega$ $\frac{1}{2\pi\sqrt{LC}} = 25$ $\frac{1}{2\pi\sqrt{LC}} = 25 \Rightarrow \frac{1}{2\pi} = 25\sqrt{LC}$ $\frac{1}{2\pi \times 25} = \sqrt{LC} \quad I = \frac{V}{R} = \frac{230}{50} = 4.6 \text{ A}$ $\therefore \sqrt{LC} = 6.36 \times 10^{-3}$ $(\sqrt{C})^2 = 6.36 \times 10^{-3}$ $\therefore \text{Capacitance, } C = 4.04 \times 10^{-5} \text{ F}$

	② voltage across capacitor $V_C = I \times \frac{1}{2\pi f C}$ $V_C = 4.6 \times \frac{1}{2\pi \times 25 \times 4.04 \times 10^{-5}} = \underline{724.86V}$ ③ Half Frequencies, $F_L = F_n - \frac{R}{4\pi L} = 25 - \frac{50}{4\pi \times 1}$ $F_L = \underline{21.02 \text{ Hz}}$ $F_H = F_n + \frac{R}{4\pi L} = 25 + \frac{50}{4\pi \times 1} = \underline{28.97 \text{ Hz}}$ ④ Q Factor, $Q = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{1}{50} \sqrt{\frac{1}{4.04 \times 10^{-5}}}$ $\cos \phi = Z/R = \underline{3.146}$ $\text{Bandwidth} = F_H - F_L = 28.97 - 21.02$ $= \underline{7.95}$
Module 5	
Q19	a) Determine the h parameters of the following network.  b) Derive the condition for symmetry and reciprocity of a two-port network in terms of transmission parameters.
Ans	a)

a)

$$V_1 = h_{11}I_1 + h_{12}V_2$$

$$I_2 = h_{21}I_1 + h_{22}V_2$$

Applying KVL mesh 1

$$V_1 - 10I_1 - 5(I_1 + I_2) = 0$$

$$V_1 - 10I_1 - 5I_1 - 5I_2 = 0$$

$$V_1 = 15I_1 + 5I_2 \quad (1)$$

with

$$V_2 - 20I_2 - 5(I_2 + I_1) = 0$$

$$V_2 = 5I_1 + 25I_2 \quad (2)$$

$$25I_2 = V_2 - 5I_1$$

$$I_2 = \frac{V_2}{25} - \frac{5I_1}{25}$$

$$V_1 = 15I_1 + 5 \left[\frac{V_2}{25} - \frac{5I_1}{25} \right]$$

$$V_1 = 15I_1 + \frac{V_2}{5} - I_1 = 14I_1 + 0.2V_2$$

$$V_1 = 14I_1 + 0.2V_2 \quad (3)$$

$$25I_2 = V_2 - 5I_1$$

$$= -5I_1 + V_2$$

$$I_2 = \frac{-5I_1}{25} + \frac{1}{25}V_2$$

$$I_2 = -0.2I_1 + 0.04V_2 \quad (4)$$

$$h_{11} = 14\Omega \quad h_{12} = 0.2 \quad h_{21} = -0.2 \quad h_{22} = 0.04\Omega^{-1}$$

b)

Transmission Parameters (ABCD Parameters)

They relate the voltage & current at i/p port to v/p & current at o/p port.

$$(V_1, I_1) = f(V_2, -I_2)$$

$$V_1 = AV_2 - BI_2$$

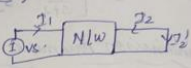
$$I_1 = CV_2 - DI_2$$

Here -ve sign is used with I_2 not for parameters B & D. In txm field o/p current is assumed to be coming out of o/p port instead of going in to the port.

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

Case 1: o/p port is o.c i.e. $I_2 = 0$

$$A = \frac{V_1}{V_2} \Big|_{I_2=0} \rightarrow \text{reverse v/p gain}$$

		$\frac{V_1}{I_1} = 0$ for admittance $V_2 = 0$ case 2: o/p port is s.c., $V_2 = 0$ $\frac{V_1}{I_1} = 0$ $\rightarrow$ transfer impedance with o/p ports c. $P = \frac{-I_1}{I_2 N_2} = 0$ reverse current gain with o/p s.c. condition for reciprocity. a) $V_1 = V_S$ $V_2 = 0$ $I_2' = -I_2$ $V_S = B I_2'$ $\frac{V_S}{I_2'} = B$  $V_2 = V_S$ $I_1' = -I_1$ $V_1 = 0$ $0 = A V_S - B I_2'$ $-I_1' = (V_S - D I_2') \Rightarrow I_2' = \frac{A}{B} V_S$ $-I_1' = C V_S - \frac{A D}{B} V_S$ $\frac{V_S}{I_1'} = \frac{B}{A D - B C}$ $\frac{V_S}{I_2'} = \frac{V_S}{I_1'} \Rightarrow \frac{V_S}{I_2'} = \frac{V_S}{I_1'} \Rightarrow B = \frac{B}{A D - B C} \Rightarrow A D - B C = 1$ <u>Condition for symmetry</u> a) when o/p port is o.c., $I_2 = 0$ $V_S = A V_2$ $I_1 = C V_2$ $\frac{V_S}{I_1} = \frac{A}{C}$ b) when i/p port is o.c., $I_1 = 0$ $C V_S = D I_2$ $\frac{V_S}{I_2} = \frac{D}{C}$ $\frac{V_S}{I_1} = \frac{V_S}{I_2} \Rightarrow \frac{A}{C} = \frac{D}{C} \Rightarrow A = D$
Q20	a) Show that Y parameters of two parallel connected two port networks is equal to the sum of their individual Y parameters. b) A two-port network is represented by the following network equations. $V_1 = 4I_1 + 2I_2$ $V_2 = 2I_1 + 6I_2$ Determine the equivalent π network.	

Ans

Given $V_1 = 4I_1 + 2I_2 \quad \text{--- (1)}$

$V_2 = 2I_1 + 6I_2 \quad \text{--- (2)}$

i.e., $V_1 = Z_{11}I_1 + Z_{12}I_2$

$V_2 = Z_{21}I_1 + Z_{22}I_2$

From (1) & (2) we get $[Z] = \begin{bmatrix} 4 & 2 \\ 2 & 6 \end{bmatrix}$

$\Delta Z = Z_{11}Z_{22} - Z_{12}Z_{21}$ $[Y] = [Z]^{-1}$
 $= 4 \times 6 - 2 \times 2$
 $= \underline{\underline{20}}$

$Y_{11} = \frac{Z_{22}}{\Delta Z} = \frac{6}{20} = 0.3$

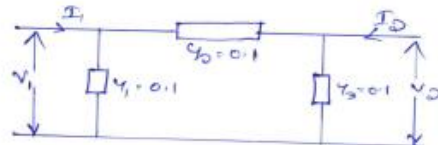
$Y_{12} = \frac{-Z_{12}}{\Delta Z} = \frac{-2}{20} = -0.1$

$Y_{21} = \frac{-Z_{21}}{\Delta Z} = \frac{-2}{20} = -0.1$

$Y_{22} = \frac{Z_{11}}{\Delta Z} = \frac{4}{20} = 0.2$

$[Y] = \begin{bmatrix} 0.3 & -0.1 \\ -0.1 & 0.2 \end{bmatrix}$

Equivalent π network



$Y_1 = Y_{11} + Y_{21} = 0.3 + (-0.1) = \underline{\underline{0.2}}$

$Y_0 = -Y_{12} = -(-0.1) = \underline{\underline{0.1}}$

$Y_2 = Y_{22} + Y_{21} = 0.2 + (-0.1) = 0.1$